

Crepant resolution conjecture : bric examples.

Reference Coates-Iritani-Jiang

Plan I introduction

II crepant transformation in bric wall crossings.

"III" Part

I : introduction

$\left\{ \begin{array}{l} \text{smooth varieties} \\ \text{(orbifolds)} \end{array} \right\} \xrightarrow{\Psi H^*} \left\{ \begin{array}{l} \text{(weak) Frobenius} \\ \text{manifolds} \end{array} \right\}$

- encodes enumerative information.
- conjecturally detects topology of X (Pukhov's Conj).

Question : Functoriality of ΨH^* under birational transformation.

basic operators:

- ① resolution of singularities
- ② blow up at smooth pt / blow down (next time??)

Today ① > { crepant resolutions }

(on): $X_1 \dashrightarrow X_2$ crepant. then $\mathbb{Q}H^*(X_1) \cong \mathbb{Q}H^*(X_2)$

Recall

• let X be Gorenstein ($\Leftrightarrow K_X$ is Cartier)

A resolution of sing. $Y \xrightarrow{f} X$ is crepant if
(f proper, Y smooth)

$$K_Y = f^* K_X$$

• examples $Y \rightarrow \mathbb{C}^2/p$ $p \in SL(2, \mathbb{C})$

• X_+ , X_- are called K -equivalent if $\exists$ X_+ X_-

such that $f_+^* K_+ = f_-^* K_-$

Rmk.

• in CRG, singular varieties has at worst quotient singularities: they are coarse moduli space of a orbifold (smooth DM stack)

* $\mathbb{Q}H^*$ can extend to orbifolds

2.1 Homological Crepant Resolution conj.

e.g.

$$\begin{array}{ccc} Y & & X \\ \parallel & & \parallel \\ \text{Tot}(\mathcal{O}_{\mathbb{P}^1}(-2)) & \rightarrow & \text{Spec}(\mathbb{C}[u, v, w] / (uv - w^2)) \\ & & \parallel \\ & & \text{Spec}(\mathbb{C}[x, y]^{\mathbb{Z}_2}) = \mathbb{C}^2 / \mathbb{Z}_2. \end{array}$$

$$\dim H^*(Y) = 2 > 1 = \dim H^*(X)$$

Betti number doesn't match!

Ruan: orbifold Chen-Ruan cohomology: $H_{\text{CR}}^*(X)$

$$\textcircled{1} \quad H_{\text{sing}}^*(X) \overset{\text{summand}}{\subset} H_{\text{CR}}^*(X)$$

$$\textcircled{2} \quad H_{\text{CR}}^*(V/G)$$

$$= \bigoplus_{[g] \in \text{conj } G} H_{\text{sing}}^{*-2\ell(g)}(V^g / C(g))$$

$$\text{Thm (Yasuda)} \quad H^*(Y, \mathbb{C}) \cong H_{\text{orb}}^*(X, \mathbb{C})$$

• as graded vector space.

• fool: motivic integration.

remark: this thm is for complete varieties. But, it's also true for some non-compact examples.

• problem: not a ring isomorphism

How to fix? ring isom after taking quantization:

Historic review:

$$\text{Ruan: } QH^{\text{small}}(Y; \mathbb{C}) \cong QH^{\text{small}}(X; \mathbb{C})$$

specify quantum parameters.

$$\text{Bryan-Graber: } F_Y \cong F_X$$

linear change of coord. + analytic continuation

— in this case, one calculates both sides and match.

Coates - Iritani - Tseng. CRG in Givental's space.

Coates - Ruan

— Recall that the Givental connection ($\hbar = \frac{1}{2}$ of Dubrovin connection)

$$\nabla_{\partial_i} = \nabla_{\partial_i}^{\text{L.C.}} - \frac{1}{2} \phi_i \star \quad \text{has basis of solutions}$$

$$s_a = L(\phi_a) = \phi_a + \sum_j \left\langle \frac{\phi_a}{2-\psi}, \phi_j \right\rangle \phi_j$$

$$\text{Let } J = \sum \langle s_e, 1 \rangle \phi^e$$

$$\text{then } \partial_i \partial_j \langle s, 1 \rangle = \frac{1}{2^2} \langle \phi_i \star \phi_j \star s, 1 \rangle$$

$$= \frac{1}{2^2} \langle C_{ij}^k \phi_k \star s, 1 \rangle = \frac{1}{2} C_{ij}^k \partial_k \langle s, 1 \rangle$$

in Givental's framework. J -functions are points in

Lagrangian cone

$$L_X \subset H_X = H_X^*(\mathbb{Z}^{-1})$$

$$L_Y \subset H_Y = H_Y^*(\mathbb{Z}^{-1})$$

CRC: • $\exists$ symplectic transformation.

$$U_{X,Y} : (H_X, L_X) \rightarrow (H_Y, L_Y)$$

sufficient, if

- $J_X = J_Y$ after change of variable
+ analytic continuation

in this version. one match quantum differential equations

Known results for this version:

(Coates - Corti - Iritani - Tseng)) some toric examples.

Coates.

Coates - Iritani - Jiang — K-equivalence in toric wall crossings

II. Crepant transformations in toric wall-crossing

Toric variety via fans

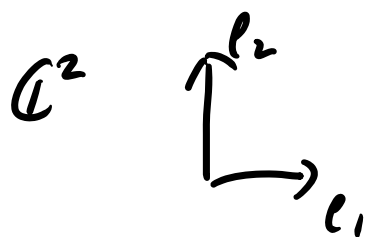
Let $N \cong \mathbb{Z}^n$ be a lattice. Σ be a collection of (convex polyhedra) cones $\Sigma = \{\sigma\}$, called a fan

then the associated toric variety X_Σ is:

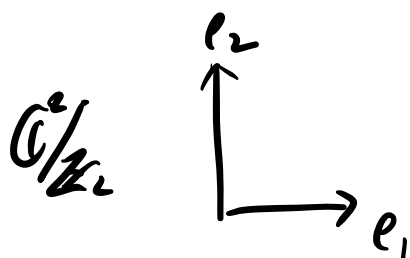
$$X_\Sigma = \bigcup_{\sigma \in \Sigma} U_\sigma \quad U_\sigma = \text{Spec } \mathbb{C}[\sigma^\vee \cap M]$$

where $M = \text{Hom}(N, \mathbb{Z})$ and $\sigma^\vee$ is dual of σ

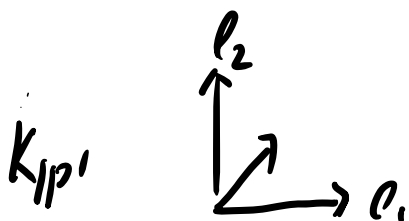
e.g. (Surfaces) $N = \mathbb{Z}^2$



$$N = \mathbb{Z}^2$$



$$N' = \mathbb{Z}^2 + \frac{1}{2}(1, 1)\mathbb{Z}$$



$$N' = \mathbb{Z}^2 + \frac{1}{2}(1, 1)\mathbb{Z}$$

crepant:

Lemma 11.4.10
in Cox Little Schenck

One can calculate
discrepancies
using combinatorics.

Toric variety via GIT quotient

Given a toric variety X_Σ . there is a standard divisor

short exact sequence.

$$0 \rightarrow M \rightarrow \mathbb{Z}^{\Sigma(1)} \xrightarrow{\deg} \mathcal{O}(X_\Sigma) \rightarrow 0$$

$m \mapsto \langle m, \sigma \rangle, \sigma \in \Sigma(1)$

By duality. $\text{Hom}(-, \mathbb{C}^*)$

$$1 \leftarrow T \leftarrow (\mathbb{C}^*)^r \leftarrow K \leftarrow 1$$

Fact: $X_\Sigma = (\mathbb{C}^r \setminus Z) //_K$

where $Z = \bigcup_{\sigma \in \Sigma} V(x_{\rho_i} \mid \rho_i \notin \sigma(1))$ is the irrelevant ideal.

On the other hand.

Given $(\mathbb{C}^*)^m \subset \mathbb{C}^r$ and a character $w: (\mathbb{C}^*)^m \rightarrow \mathbb{C}$

The GIT quotient $\mathbb{C}^r //_{(\mathbb{C}^*)^m} = (\mathbb{C}^r - \underline{Z(w)}) //_{(\mathbb{C}^*)^m}$

Semi-stable locus

rank the weights of $(\mathbb{C}^*)^m \subset \mathbb{C}^r$ give a wall & chamber structure. called secondary fan / GKZ fan

VAT for toric varieties

1. Let $V = \mathbb{C}^m$ $(\mathbb{C}^*)^r = K \cong \mathbb{C}^r$

$$V = \bigoplus_{i=1}^m V_{D_i} \quad \text{where}$$

$$D_i \in \text{Hom}(K, \mathbb{C}^*) = L \cong \mathbb{Z}^r$$

$$\text{then } \text{Cone}(X_i) \in \text{Hom}(T, \mathbb{C}^*) \otimes_{\mathbb{Z}} \mathbb{R} \\ \cong \mathbb{R}^r$$

gives walls in $\mathbb{R}^r$

denote it by C_p

Given any other $w \in \text{Hom}(K, \mathbb{C}^*)$

$$\text{Let } R^w = \bigoplus_{i=0}^{\infty} R_i^w$$

$$R_i^w = \{x^I \mid (p(t)x)^I = w(t)x^I\}$$

$$X^w = \mathbb{Z}(R_+^w)$$

2. Semi-stable criterion

$$\text{Let } \tilde{D} = \{D \mid v_D \neq 0\}$$

$$\forall x \in V. \quad x = \sum x_D \quad \text{denote}$$

$$\tilde{D}(x) = \{D \in \tilde{D} \mid x_D \neq 0\}$$

$$\text{Prop } x \in X^{ss}(w)$$

$$\Leftrightarrow w \in \text{cone } \tilde{D}(x).$$

$$\text{Prop. (Coates - Britani - Jiang \S 5.1)}$$

$$\text{let } X_+ = \vee_{w_+} K$$

$$X_- = \vee_{w_-} K$$

related by crossing a wall W

then $X_+ \dashrightarrow X_-$ crepant if (only if?)

$\sum D_i \in W$ (sum with multiplicities)

II. Example 1: $X = [\mathbb{C}^N / \mathbb{Z}_N]$ $Y = K_{\mathbb{P}^{N-1}}$
 $= \text{Tot}(\mathcal{O}_{\mathbb{P}^N}(-N))$

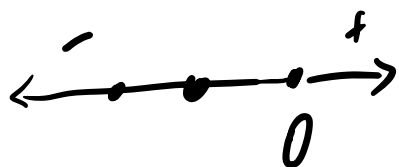
X and Y as GIT quotients

Consider the $\mathbb{C}^*$ -action on $\mathbb{C}^{N+1}$ by:

$$t \begin{pmatrix} x_1 \\ \vdots \\ x_n \\ y \end{pmatrix} = \begin{pmatrix} tx_1 \\ \vdots \\ tx_n \\ t^{-n}y \end{pmatrix}$$

the weights 1 and $-N$ generates a fan in

$$\text{Hom}(\mathbb{C}^*, \mathbb{C}^*) \otimes_{\mathbb{Z}} \mathbb{R} \cong \mathbb{R}$$



with the standard lattice structure $\mathbb{Z} \subset \mathbb{R}$

the two chambers gives rise to two GIT quotients.

$$X^{ss}(\pm) // \mathbb{C}^*$$

$$\text{where } X^{ss}(-) = \mathbb{C}^N \times (\mathbb{C} - \{0\})$$

$$X^{ss}(+) = (\mathbb{C}^N - \{0\}) \times \mathbb{C}$$

from this one get:

$$X = X^{ss}(-) //_{\mathbb{C}^\times} = \mathbb{C}^N / \mathbb{Z}_N$$

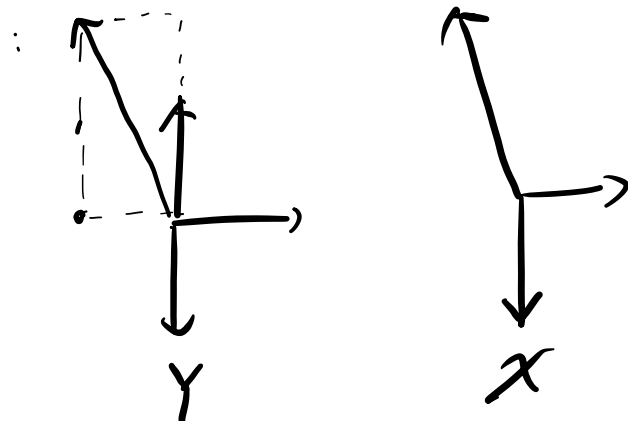
$$Y = X^{ss}(+) //_{\mathbb{C}^\times} = K_{\mathbb{P}^{N-1}}.$$

$$\dim H^*(K_{\mathbb{P}^{N-1}}) = \dim H^*(\mathbb{P}^{N-1}) = N$$

$$\dim H_{CR}^*(\mathbb{C}^N / \mathbb{Z}_N) = \# \text{conj } \mathbb{Z} / N\mathbb{Z} = N.$$

Example 2 $Y = \mathbb{A}^2$ $X = \mathbb{P}(1,1,2)$

$\mathbb{A}^2$ and $\mathbb{P}(1,1,2)$
as toric varieties



$\mathbb{A}^2$ as a GIT quotient: $(\mathbb{C}^2 \setminus \{0\} \times \mathbb{C}^2 \setminus \{0\}) //_{(\mathbb{C}^*)^2}$

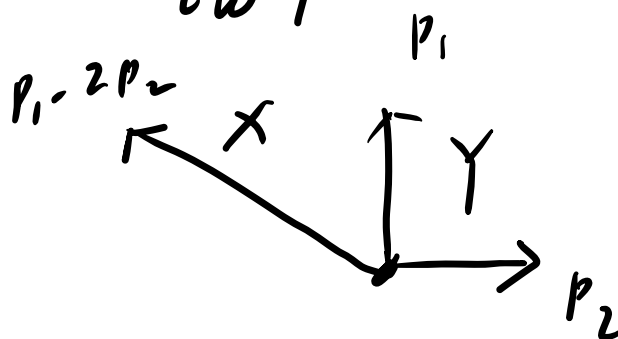
$$\begin{pmatrix} 1 & 1 & -2 & 0 \\ 0 & 0 & 1 & 1 \end{pmatrix} \quad \begin{pmatrix} 1 & 0 \\ -1 & 2 \\ 0 & 1 \\ 0 & -1 \end{pmatrix}$$

$$0 \rightarrow \mathbb{Z}^2 \rightarrow \mathbb{Z}^4 \rightarrow \mathbb{Z}^2 \rightarrow 0$$

where $(\mathbb{C}^*)^2$ is given by:

$$(s, t) \begin{pmatrix} x \\ y \\ z \\ w \end{pmatrix} = \begin{pmatrix} sx \\ sy \\ s^2tz \\ tw \end{pmatrix}$$

secondary fan:



$\mathbb{F}^2$: $\mathbb{P}^1$ bundle over $\mathbb{P}^1$

$$\dim H^*(\mathbb{F}^2) = 2 \times 2 = 4$$

$$H^*_{CR}(\mathbb{P}(1,1,2))$$

$$= H^*(\mathbb{P}(1,1,2)) \oplus H^{*-1}(\mathbb{P}(2))$$

$$\dim H^*_{CR}(\mathbb{P}(1,1,2)) = 3 + 1 = 4$$

—

$\mathcal{M}_B$:

$$y_1 = \hat{y}_1 \hat{y}_2$$

$$y_2 = \hat{y}_2^{-2}$$

III Proof of CRC by Mirror Symmetry and global D-module

General Strategy:

1. Toric Mirror Theorem

$$I(q, z) = J(\tau(q), z)$$

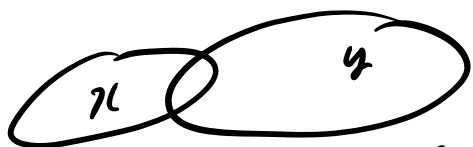
$$\triangleright^{\text{Givental}} J = 0$$

$$\Downarrow$$

$$\triangleright^{\text{GM}} I = 0$$

2. global D-module (Gelfand-Kapranov-Zelovinskij system) associated to the secondary fan.

$$\Pi_x|_{uv} = \Pi_y|_{uv}$$



$$\Pi_x I_x^0 \quad \Pi_y I_y$$

$$= M_B$$

$$J = z + \tau + \ll \frac{\phi^e}{z - \psi} \gg \phi^e$$

12 γ :

$$1 + (1p^2) = G[p] / \langle p^2 \rangle$$

$$\tau = t_0 1 + t_1 p + t_2 p^2$$

$$q = e^{t_1}$$

$$1 + (K_{1p^2}) = G(\lambda)[p] / \langle p^2 \rangle$$

Given that: (see also Cox & Katz (10.391))

$$J_{1p^2}^{\text{small}}(q, z) = z q^{p/2} \sum_{d \geq 0} \frac{q^d}{\prod_{0 \leq m \leq d} (p + mz)^3}$$

$$q = e^{t_1}$$

$$\langle \gamma_{K_{1p^2}} \rangle = \langle \gamma_{1p^2}^{(0(-3), tw)} \rangle$$

$$I_{K_{1p^2}}(y, z) = I_{1p^2}^{(0(-3), tw)}(q, z)$$

$$= z \sum_{d \geq 0} \frac{\prod_{-3d \leq m \leq 0} (\lambda - 3p + mz)}{\prod_{0 \leq m \leq d} (p + mz)^3} y^{d+p/2}$$

Mirror Symmetry

$$J_Y(y, z) = e^{\lambda f(y)/z} I_Y(y, z)$$

$$q = \exp(\lambda f(y))$$

$$f(y) = \sum_{d \geq 0} \frac{(3d-1)!}{(d!)^3} (1-y)^d$$

$$2) \left[\mathbb{Q}^3 / \mathbb{Z}_3 \right] \rightarrow \left[\frac{*}{\mathbb{Z}_3} \right] \quad \tau' = \tau^0 \mathbb{1}_0 + \tau^1 \mathbb{1}_{\frac{1}{3}} + \tau^2 \mathbb{1}_{\frac{2}{3}}$$

{ //

$$\langle \quad \rangle_{[\mathbb{Q}^3 / \mathbb{Z}_3]} = \langle \quad \rangle_{\substack{*, \text{tw} \\ [\frac{*}{\mathbb{Z}_3}]}}$$

$$I_X^{(\lambda)} = z x^{-\lambda/2} \sum_{d \geq 0} \frac{x^d}{d! z^d} \prod_{\substack{b: 0 \leq b \leq \frac{1}{3} \\ \text{frac } b = \text{frac } \frac{1}{3}}} \left(\frac{\lambda}{3} - bz \right)^3 \mathbb{1}_{\text{frac}(\frac{\lambda}{3})}$$

frac b = frac \frac{1}{3}

Mirror Symmetry

$$J_X^{\text{big}}(\tau' \parallel_{1/3}, z) = x^{\lambda/2} I_X(\lambda, z)$$

$$\tau' = \sum_{m \geq 0} (-1)^m \frac{x^{3m+1}}{(3m+1)!} \frac{\Gamma(m + \frac{1}{3})^3}{\Gamma(\frac{1}{3})^3}$$

D-module: $y = x^{-3}$, $D_y = z y \frac{\partial}{\partial y}$, $D_x = z x \frac{\partial}{\partial x}$

$$D_y^3 I_Y = y (\lambda - 3 D_y) (\lambda - 3 D_y - z) (\lambda - 3 D_y - 2z) I_Y$$

$$D_x^3 I_X = -27 x^{-3} (\lambda + D_x) (\lambda + D_x - z) (\lambda + D_x - 2z) I_X$$

Analytic continuation: λ : equivariant parameter

$$\tilde{I}_Y = M(\lambda, z) I_X$$

3. Some toric birational geometry.

Let w be a element in a chamber.

(Oates - Iritani - Jiang's notation:

$$A_w = \{ I \in \{1, \dots, m\} : w \in \text{cone}(I) \}$$

$$\text{then } U_w = X^{ss}(w) = \bigcup_{I \in A_w} (\mathbb{C}^*)^I \times \mathbb{C}^{\bar{I}}$$

$$C_w = \bigcap_{I \in A_w} \angle I \quad \text{is called the extended ample cone}$$

$$\text{let } S = \{ i \in \{1, \dots, m\} : \{1, \dots, m\} \setminus \{i\} \notin A_w \}$$

$$\text{then } A_w = \{ I \sqcup S : I \in A_{w'} \}$$

$$U_w = U_{w'} \times (\mathbb{C}^*)^{|S|}$$

$$\text{let } \mathcal{L}_w' = \bigcap_{I \in A_w'} \mathcal{L}_I' \subset \mathcal{L}^\vee \otimes \mathbb{R} / \sum_{i \in S} \mathbb{R} D_i$$

then

$$\mathcal{L}_w = \mathcal{L}_w' \times \left(\sum_{i \in S} \mathbb{R} D_i \right)$$

$$\subset H^2(X_w; \mathbb{R}) \times \bigoplus_{i \in S} \mathbb{R} D_i$$

$$NE(X_w) = \mathcal{L}_w'^\vee \subset H^2(X_w; \mathbb{R})$$